SCHOOL OF COMPUTER TECHNOLOGY

AASD 4008
Big Data Tools and Technologies II

Data Warehouse ?

data warehouse is relational in nature.

The structure or schema is modeled or predefined by business and product requirements that are curated, conformed, and optimized for SQL query operations.

data warehouse stores data that has been treated and transformed with a specific purpose in mind, which can then be used to source analytic or operational reporting. This makes data warehouses ideal for producing more standardized forms of BI analysis, or for serving a business use case that has already been defined.

** https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-a-data-lake

What is a Data Lake ?

A data lake is a centralized repository that ingests and stores large volumes of data in its original form. The data can then be processed and used as a basis for a variety of analytic needs. Due to its open, scalable architecture, a data lake can accommodate all types of data from any source, from structured (database tables, Excel sheets) to semi-structured (XML files, webpages) to unstructured (images, audio files, tweets), all without sacrificing fidelity.

The data files are typically stored in

- staged zones—raw
- cleansed
- Curated

So now different types of users may use the data in its various forms to meet their needs. Data lakes provide core data consistency across a variety of applications, powering big data analytics, machine learning, predictive analytics, and other forms of intelligent action.

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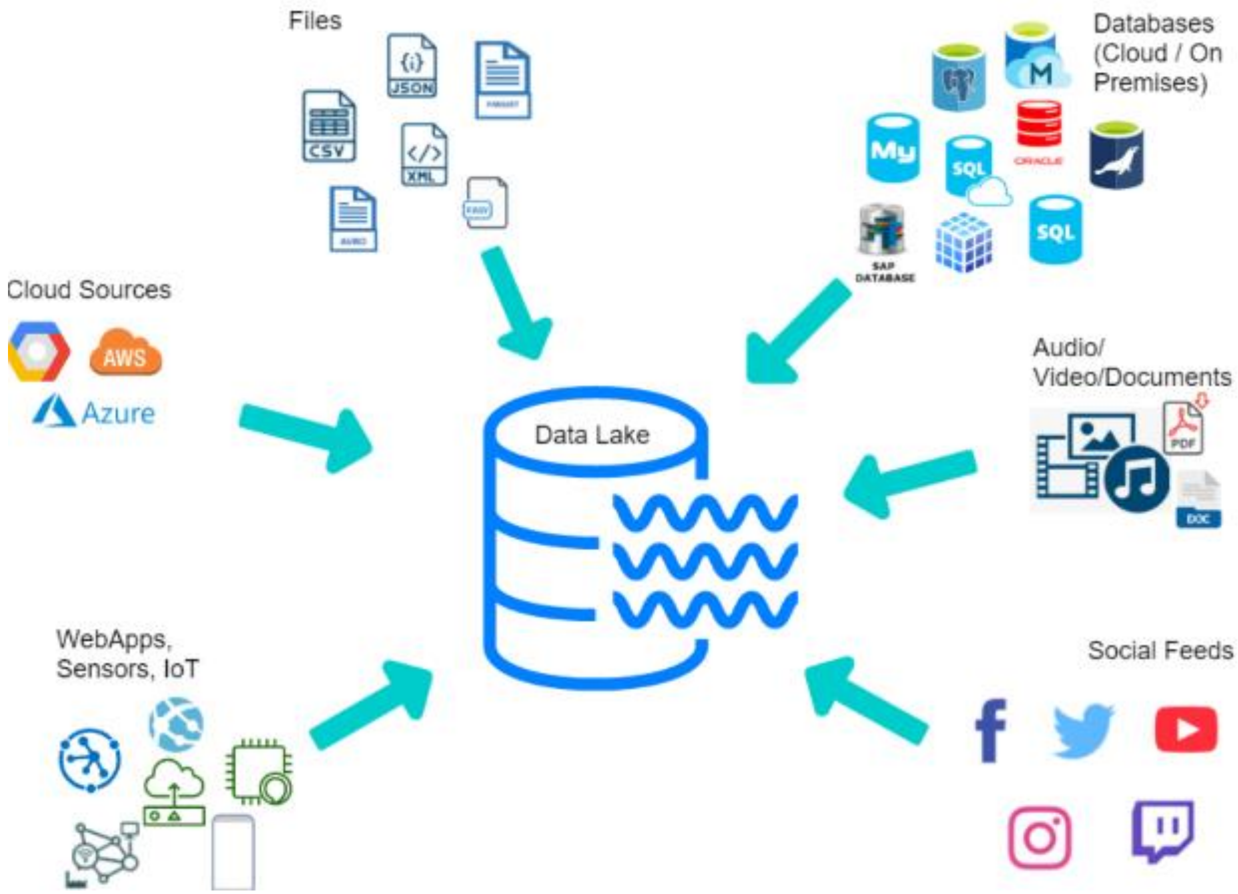
What is a Data Lake ?



** https://vitalflux.com/data-lake-design-principles-best-practices/

Data Lake vs Data Warehouse Similarities

data lakes and data warehouses are similar in their purpose , both store and process data, each have their own specialties, and therefore their own use cases.

Its common for an enterprise-level organization to include a data lake and a data warehouse in their analytics ecosystem.

** https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-a-data-lake

Data Lake vs Data Warehouse

	Data lake	Date warehouse
Type	Structured, semi-structured and unstructured	Structured
Nature	Relational , non – relational	Relational
Schema	Schema on read	Schema on write
Format	Raw , unfiltered, processed and curated	Processed, vetted
Sources	Big data, Iot, social medias, streaming data	Applications, business, transactional data , batch reporting
Scalability	Easy to scale at low cost	Difficult and expensive to scale
Users	Data scientists, data engineers	Data scientists, data engineers, DW professionals and business analysts

** https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-a-data-lake

Journey to Data Lakehouse

Datebases are structured and organized storage for efficient data storage and retrieval. They are used for OLTP and optimized for read and write . Not good for analytics and historical data analysis. Datawarehouse answers the call.

for a while data warehouses allow businesses to organize historical datasets for use in business intelligence (BI) and analytics, they quickly become more cost-intensive as datasets grow because of the combined use of compute and storage resources. Additionally, data warehouses can't handle the varied nature of data (structured, unstructured, and semi-structured) seen today.

Data lakes answered this problem with their low-cost of storage and ability to handle data in various formats. Furthermore, their ability to manage data in multiple formats makes data lake architecture applicable to different use cases but data lakes have limitations, too; their storage of data as files without a defined schema makes it challenging to perform critical data management to enable ETL transactions and leads to the creation of orphaned data. This is where the need for Data lakehouse

** https://streamsets.com/blog/delta-lake-architecture-bridge-between-data-lakes-data-warehouses/

Journey to Data Lakehouse continued

data lakehouse is a relatively new concept that combines the best of data lakes and data warehouses. It aims to provide the flexibility of a data lake for storing diverse, raw data while adding structured query capabilities typically associated with data warehouses. This hybrid approach often uses technologies like Delta Lake, which adds ACID (Atomicity, Consistency, Isolation, Durability) transactions and schema enforcement to the data lake.

** https://streamsets.com/blog/delta-lake-architecture-bridge-between-data-lakes-data-warehouses/

Azure Data lakehouse

A data lakehouse is an open standards-based storage solution that is multifaceted in nature. The workload can seamlessly operate on top of the data lake without having to duplicate

Data lakehouses address the challenges of traditional data lakes by adding a Delta Lake storage layer directly on top of the cloud data lake.

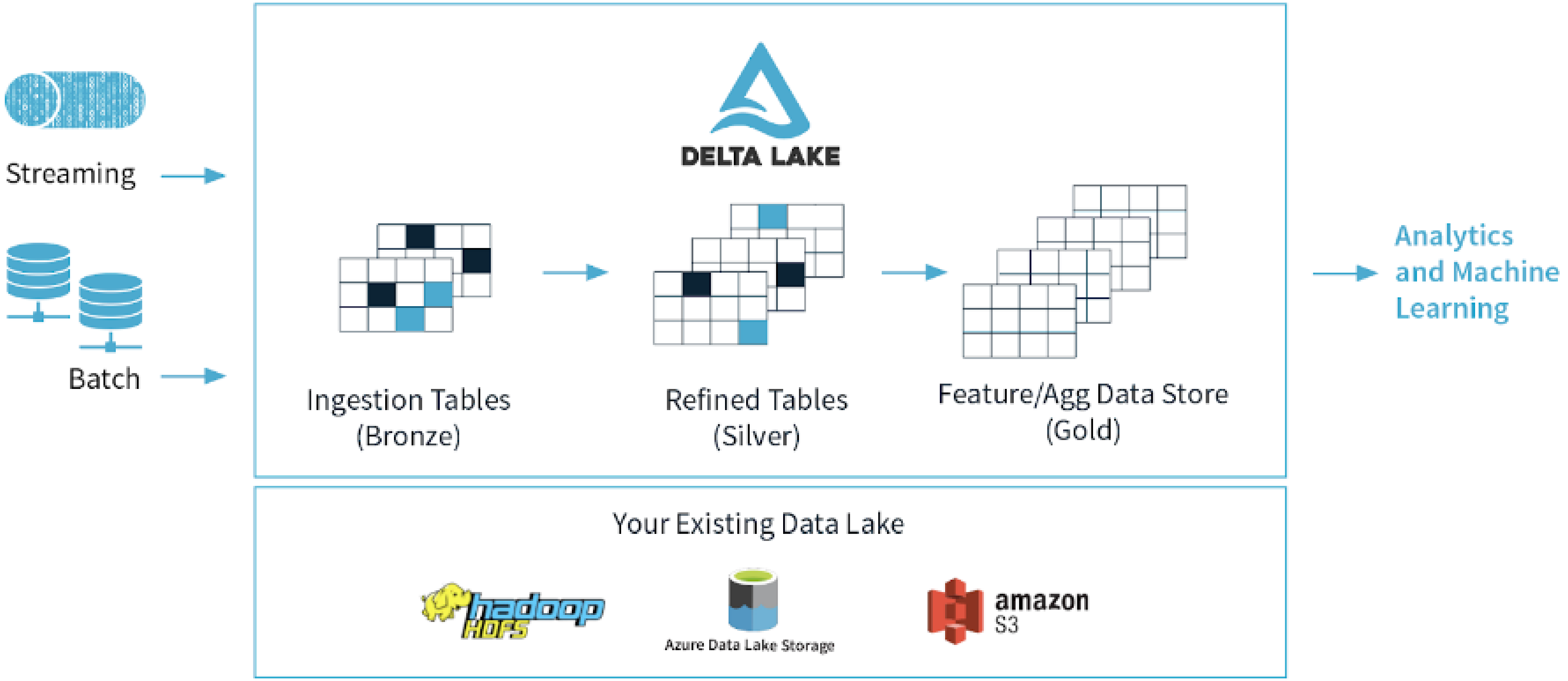
Delta lake provides a flexible analytic architecture that can handle ACID (atomicity, consistency, isolation, and durability) transactions for data reliability, streaming integrations, and advanced features like data versioning and schema enforcement.

Data Lake + Delta Lake Storage -> Data Lakehouse

DELTA LAKE

** https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-a-data-lake

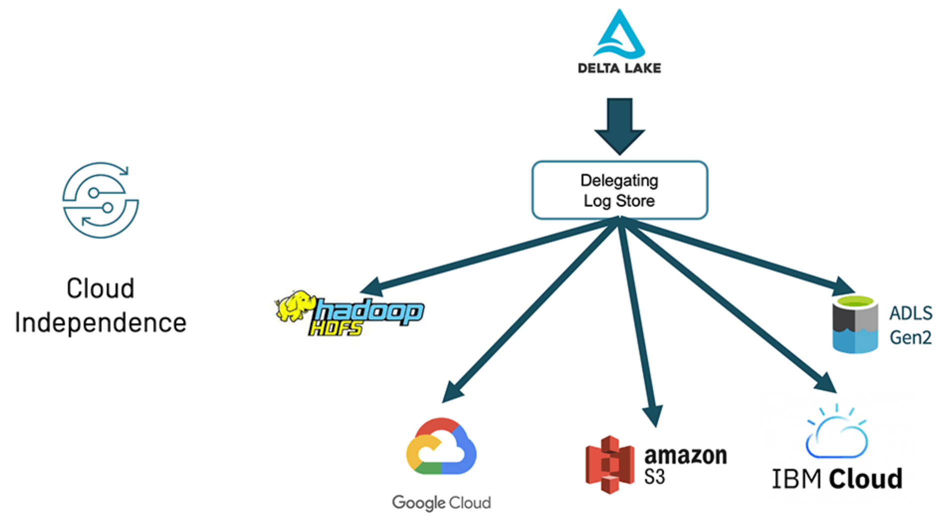
Delta Lake



** https://www.databricks.com/blog/2021/12/01/the-foundation-of-your-lakehouse-starts-with-delta-lake.html

Delta Lake – cloud independence

Out of the box, Delta Lake has always worked with a variety of storage systems – Hadoop HDFS, Amazon S3, Azure Data Lake Storage (ADLS) Gen2 – though the cluster would previously be specific for one storage system.



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Data Lake vs data lakehouse

	Data lake	Data lakehouse
Type	Structured, semi-structured and unstructured	Same as datalake
Nature	Relational , non – relational	Same as datalake
Schema	Schema on read	Schema on read Schema on write
Format	Raw , unfiltered, processed, curated	Same as datalake + delta format files
Sources	Big data, Iot, social medias, streaming data	Applications, business, transactional data , batch reporting
Scalability	Easy to scale at low cost	Same as datalake
Users	Data scientists, data engineers	Same as datalake + DW professionals and business analysts

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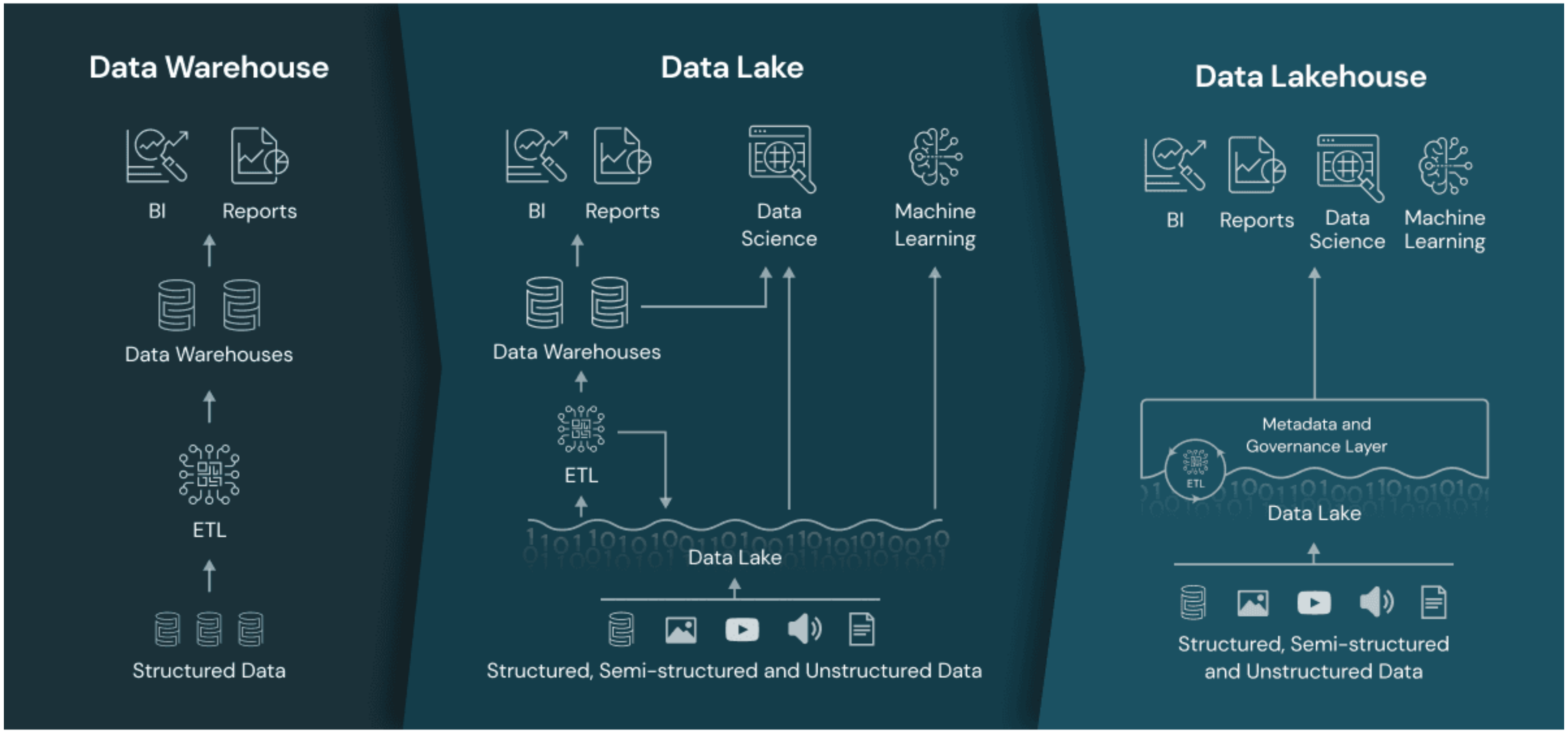
Bring them all together



An ETL is required to read from datalake for BI/Reports , however, with data Lakehouse, the ETL necessity is mostly replaced with Delta Lake

** https://www.databricks.com/de/glossary/data-lakehouse
** https://www.databricks.com/blog/2020/01/30/what-is-a-data-lakehouse.html

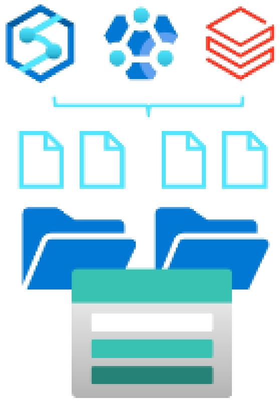
Azure Data Lake Gen 2

Many BI solutions have lost out on opportunities to store unstructured data due to cost and complexity in these types of data in databases and data lake become a common solution to this problem by proving a file based storage usually in distributed file system that supports high scalability for massive volume of data.

Azure Data Lake Storage Gen2 provides a cloud-based solution for data lake storage in Microsoft Azure, and underpins many large-scale analytics solutions built on Azure.

Blob storage that

1. Organize and store data into hierarchy of directories and subdirectories so data requires less computational resource
2. Provide file-based storage
3. Data can be consumed using Apache spark !
4. Combines filesystem and storage platform to process and store data on exabyte scale handling several Hundreds of gigabyte throughput



** https://learn.microsoft.com/en-us/training/modules/introduction-to-azure-data-lake-storage/1-introduction

Stages of processing big data

Data lakes have a fundamental role in a wide range of big data architectures. These architectures can involve the creation of

- An enterprise data warehouse.
- Advanced analytics against big data.
- A real-time analytical solution.

Four Stages of processing big data solutions

1. Ingest
2. Store
3. Prep and train
4. Model and Serve

** https://learn.microsoft.com/en-us/training/modules/introduction-to-azure-data-lake-storage/5-stages-for-processing-big-data

Stages of processing big data Continued

Ingest

The ingestion phase identifies the technology and processes that are used to acquire the source data. The technology that is used will vary depending on the frequency that the data is transferred.

1. Batch movement : Pipelines in Azure Synapse Analytics and Azure Data Factory etc…
2. real-time ingestion : Apache Kafka for HDInsight or Stream Analytics etc..

Store

The store phase identifies where the ingested data should be placed. Azure Data Lake Storage Gen2 provides a secure and scalable storage solution that is compatible with commonly used big data processing technologies.

Prep and train

Identifies the technologies that are used to perform data preparation and model training and scoring for machine learning solutions. Common technologies that are used in this phase are Azure Synapse Analytics, Azure Databricks, Azure HDInsight, and Azure Machine Learning.

Model and serve

involves the technologies that will present the data to users. These technologies can include visualization tools such as Microsoft Power BI, or analytical data stores such as Azure Synapse Analytics.

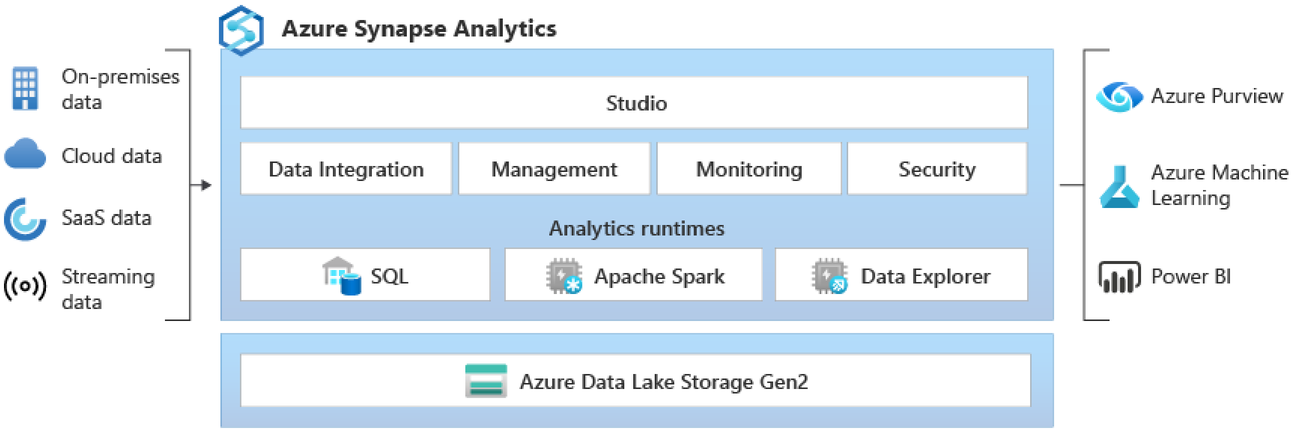
** https://learn.microsoft.com/en-us/training/modules/introduction-to-azure-data-lake-storage/5-stages-for-processing-big-data

What is Azure Synapse Analytics

Azure Synapse is an enterprise analytics service that accelerates time to insight across data warehouses and big data systems.

Azure Synapse brings together the best of

- **SQL** technologies used in enterprise data warehousing
- **Spark** technologies used for big data
- **Data Explorer** for log and time series analytics
- **Pipelines** for data integration and ETL/ELT, and
- deep integration with other Azure services such as **Power BI**, **CosmosDB**, and **AzureML**.



Use of SQL – Synapse SQL

Synapse SQL is a distributed query system for T-SQL that enables data warehousing and data virtualization scenarios and extends T-SQL to address streaming and machine learning scenarios.

Synapse SQL offers both **serverless** and **dedicated** resource models. For predictable performance and cost

- create dedicated SQL pools to reserve processing power for data stored in SQL tables.
- For unplanned or bursty workloads, use the always-available, serverless SQL endpoint.

Synapse SQL use built-in **streaming** capabilities to land data from cloud data sources into SQL tables

Integrate AI with SQL by using **machine learning** models

Azure Synapse SQL architecture

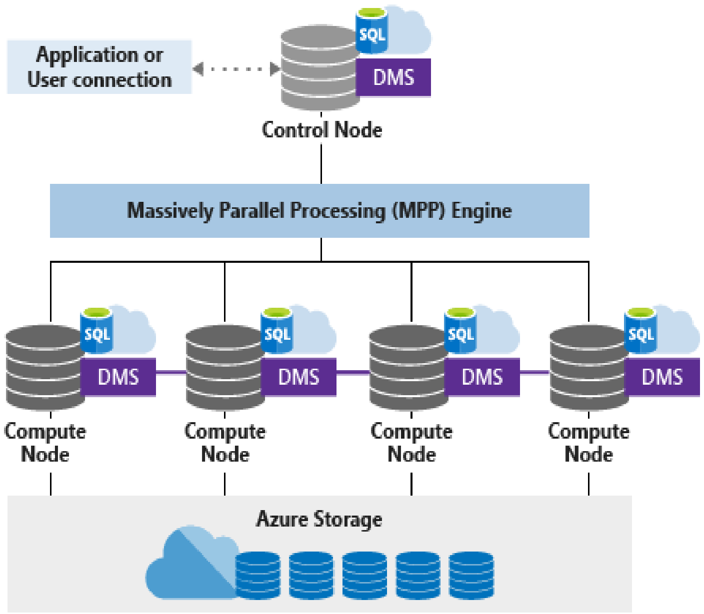
Synapse SQL uses a scale-out architecture to distribute computational processing of data across multiple nodes.
Compute is separate from storage, which enables you to scale compute independently of the data in your system

For dedicated SQL pool, the unit of scale is an abstraction of compute power that is known as a data warehouse unit.

For serverless SQL pool, being serverless, scaling is done automatically to accommodate query resource requirements. As topology changes over time by adding, removing nodes or failovers, it adapts to changes and makes sure your query has enough resources and finishes successfully.

Azure Synapse SQL architecture

Dedicated SQL pool



Serverless SQL pool